

Supplementary Materials

Neuroelastic Resonance Instability: The SymC Theory of Addiction
Technical Appendices, Measurement Protocols, and Extended Derivations

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1 Substrate Inheritance

1.1 The Universal Boundary

The critical damping boundary $\chi = 1$ is not imposed on biological systems from outside; it emerges from fundamental physics. In any second-order dissipative system:

$$\ddot{x} + \gamma\dot{x} + \omega^2x = 0 \quad (1)$$

the discriminant $\Delta = \gamma^2 - 4\omega^2$ governs response character:

- $\Delta < 0$ ($\chi < 1$): Underdamped, oscillatory approach to equilibrium
- $\Delta = 0$ ($\chi = 1$): Critical damping, fastest non-oscillatory decay
- $\Delta > 0$ ($\chi > 1$): Overdamped, sluggish exponential decay

At $\chi = 1$, information efficiency is maximized: the system extracts maximum signal from perturbations while dissipating minimum energy per bit processed.

1.2 Quantum Origins

In open quantum systems governed by the Lindblad master equation:

$$\dot{\rho} = -i[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right) \quad (2)$$

the competition between coherent evolution (ω) and dissipation (γ) produces the same boundary structure. Numerical simulations suggest an effective critical boundary at $\chi_Q \approx 1.0$, with a narrow transition band (± 0.05) in typical parameter regimes.

1.3 Neural Inheritance

Neural systems inherit χ -constraints from their biophysical substrates:

- **Ion channels:** GABA_A receptor kinetics operate near the critically damped regime
- **Synaptic transmission:** Vesicle release/reuptake dynamics avoid both ringing and sluggish decay
- **Network oscillations:** Cortical gamma rhythms behave as near-critically damped oscillators

The macroscopic reward-control system cannot achieve χ values that its substrates cannot support. Stable systems are built from stable substrates: $\chi \approx 1$ propagates upward through the organizational hierarchy.

2 Multi-Mode Coupling Mathematics

2.1 The Coupling Matrix

The subsystems do not evolve independently. Define the state vector:

$$\vec{\chi}(t) = \begin{bmatrix} \chi_{\text{reward}} \\ \chi_{\text{control}} \\ \chi_{\text{stress}} \\ \chi_{\text{habit}} \\ \chi_{\text{interoception}} \\ \chi_{\text{environment}} \end{bmatrix} \quad (3)$$

The coupling matrix captures cross-system interactions:

$$\mathbf{C} = \begin{pmatrix} 1 & c_{rc} & c_{rs} & c_{rh} & c_{ri} & c_{re} \\ c_{cr} & 1 & c_{cs} & c_{ch} & c_{ci} & c_{ce} \\ c_{sr} & c_{sc} & 1 & c_{sh} & c_{si} & c_{se} \\ c_{hr} & c_{hc} & c_{hs} & 1 & c_{hi} & c_{he} \\ c_{ir} & c_{ic} & c_{is} & c_{ih} & 1 & c_{ie} \\ c_{er} & c_{ec} & c_{es} & c_{eh} & c_{ei} & 1 \end{pmatrix} \quad (4)$$

Key coupling terms:

- c_{rs} : Stress $\rightarrow$ Reward coupling (stress amplifies cue salience)
- c_{cs} : Stress $\rightarrow$ Control coupling (stress degrades PFC function)
- c_{rh} : Habit $\rightarrow$ Reward coupling (automaticity bypasses control)
- c_{er} : Environment $\rightarrow$ Reward coupling (cue-triggered activation)

2.2 Coupled Dynamics

The full system evolves as:

$$\dot{\vec{\chi}} = \mathbf{A}\vec{\chi} + \mathbf{C}(\vec{\chi} - \vec{\chi}^*) + \mathbf{B}\vec{u}(t) + \vec{\xi}(t) \quad (5)$$

where:

- $\mathbf{A}$: Intrinsic dynamics matrix
- $\vec{\chi}^* = [0.9, 0.9, 0.9, 0.9, 0.9, 0.9]^T$: Target attractor
- $\vec{u}(t)$: Control inputs (therapy, medication)
- $\vec{\xi}(t)$: Stochastic perturbations

2.3 Phase Divergence Criterion

Relapse risk increases when subsystems decouple:

$$\text{Divergence Index: } D(t) = \sqrt{\sum_{i < j} w_{ij} (\chi_i - \chi_j)^2} \quad (6)$$

In simulations, $D > D_{\text{crit}} \approx 0.4$ is associated with high relapse risk within 72 hours.

The specific reward-control divergence:

$$|\chi_{\text{reward}}(t) - \chi_{\text{control}}(t)| > \delta_{\text{crit}} \approx 0.3 \quad (7)$$

captures the “knowing I shouldn’t but doing it anyway” state: phase-lock failure between drive and inhibition.

3 χ -Curvature and Trajectory Analysis

3.1 Curvature as Relapse Predictor

Relapse correlates not with χ -value but with trajectory curvature:

$$\kappa(t) = \frac{|\ddot{\chi}|}{[1 + \dot{\chi}^2]^{3/2}} \quad (8)$$

High curvature indicates rapid approach to boundary crossing. A patient with $\chi = 0.75$ and $\kappa = 0.1$ (gentle drift) is safer than one with $\chi = 0.82$ and $\kappa = 0.8$ (rapid destabilization).

3.2 Early Warning Threshold

Preliminary modeling suggests a threshold:

$$\kappa(t) > \kappa_{\text{warn}} \approx 0.5 \quad (9)$$

triggers intervention protocol before χ crosses 0.75.

3.3 Trajectory Classification

Four trajectory types correspond to the failure modes:

- **Type A:** Monotonic descent, $\dot{\chi} < 0$, κ low
- **Type B:** Spiral/chaotic, high κ variance
- **Type C:** Step functions at environmental transitions
- **Type D:** Fragmented, multi-basin switching

4 Environmental Impedance and Boundary Reflections

The following expressions are schematic, indicating that both cue density and context stiffness contribute to an effective impedance rather than specifying unique biophysical formulas.

4.1 Impedance Matching Theory

The environment has characteristic impedance:

$$Z_{\text{env}} = \sqrt{\rho_{\text{cue}} \cdot G'_{\text{context}}} \quad (10)$$

where ρ_{cue} is cue density and G'_{context} is context stiffness.

The reward system has its own impedance:

$$Z_{\text{reward}} = \sqrt{\rho_{\text{neural}} \cdot G'_{\text{circuit}}} \quad (11)$$

4.2 Reflection Coefficient

Energy transfer at the boundary:

$$R = \frac{Z_{\text{env}} - Z_{\text{reward}}}{Z_{\text{env}} + Z_{\text{reward}}} \quad (12)$$

When $Z_{\text{env}} \approx Z_{\text{reward}}$ (matched load):

- $R \rightarrow 0$: Maximum energy transfer
- Cues drive directly into reward system
- Relapse is instantaneous

When $Z_{\text{env}} \neq Z_{\text{reward}}$ (mismatched):

- $|R| \rightarrow 1$: Energy reflects
- Cues bounce off, reduced penetration
- Protection from environmental triggers

4.3 Therapeutic Impedance Mismatch

Treatment strategy: *increase* mismatch to protect against environmental triggers.

- Reduce Z_{env} : Remove cues, change context
- Increase Z_{reward} : Build γ , reduce circuit sensitivity
- Create reflection rather than transmission

5 Measurement Protocols for $\chi(t)$ Extraction

5.1 Lorentzian PSD Method

For band-limited signals $x(t)$:

1. Compute power spectral density via Welch method
2. Fit dominant peak with Lorentzian:

$$P(f) = \frac{A(\Gamma/2)^2}{(f - f_0)^2 + (\Gamma/2)^2} + C \quad (13)$$

3. Extract parameters:
 - $f_0 \rightarrow \omega = 2\pi f_0$ (dominant frequency)
 - Γ (half-power bandwidth)
4. Compute: $\hat{\chi} = \Gamma/(2f_0)$

Applications: EEG beta bursts, HRV spectral peaks, craving rhythmicity.

5.2 Hilbert Envelope Method

For transient/burst signals:

1. Bandpass filter to frequency band of interest
2. Compute analytic signal: $z(t) = x(t) + i\mathcal{H}[x(t)]$
3. Extract envelope: $A(t) = |z(t)|$
4. Fit exponential decay: $A(t) \approx A_0 e^{-\gamma t/2}$
5. Extract instantaneous frequency: $\omega(t) = \dot{\phi}(t)$
6. Compute: $\chi(t) = \gamma(t)/(2|\omega(t)|)$

Applications: Cue-evoked responses, craving bursts, withdrawal oscillations.

5.3 Behavioral Autocorrelation Method

For choice sequences or behavioral time series:

1. Compute autocorrelation $R(\tau)$ of inter-response times
2. Fit exponential envelope: $R(\tau) \approx R_0 e^{-\gamma \tau/2}$
3. Estimate ω from peak inter-event timing
4. Compute: $\hat{\chi} = \gamma/(2\omega)$

Applications: Go/no-go variability, craving report periodicity, decision noise.

5.4 N4SID Multimodal Fusion

Gold standard for combined signals:

1. Assemble measurement vector: $\vec{y}(t) = [\text{EEG}, \text{HRV}, \text{behavior}, \text{cortisol}]^T$
2. Build Hankel matrix from time series
3. Perform subspace state-space identification (N4SID)
4. Extract state matrix $\mathbf{A}$
5. Eigenvalues yield: $\lambda_{\pm} = -\gamma \pm i\omega$
6. Compute: $\chi = \gamma/(2|\omega|)$

Advantages: Integrates all available information, handles noise, provides confidence intervals.

6 Neurobiological Mapping

6.1 ω Components (Drive)

Table 1: Neural substrates of reward drive (ω)

Component	Substrate	Measurement	Timescale
Phasic DA	VTA $\rightarrow$ NAc	fMRI NAc, voltammetry	100ms–1s
Tonic DA	Baseline firing	PET raclopride	Minutes–hours
Glutamate	PFC $\rightarrow$ striatum	Glutamate MRS	Minutes
Cue salience	Insula, ACC	EEG theta, fMRI	Seconds
Craving intensity	Distributed	Self-report, HRV	Variable

6.2 γ Components (Damping)

Table 2: Neural substrates of inhibitory control (γ)

Component	Substrate	Measurement	Timescale
D2R indirect	D2 receptors	PET raclopride	Hours
PFC control	dIPFC $\rightarrow$ striatum	fMRI connectivity	Seconds
GABA interneurons	NAc local	GABA-MRS	Minutes
Habenula	LHb $\rightarrow$ VTA	fMRI activation	Seconds
Beta oscillations	Frontal networks	EEG frontal β	100ms

6.3 Stress as Dual-Axis Perturbation

Stress attacks both variables simultaneously:

$$\text{Cortisol} \rightarrow \gamma_{\text{PFC}} \downarrow \text{ (blunts prefrontal function)} \quad (14)$$

$$\text{Norepinephrine} \rightarrow \omega_{\text{salience}} \uparrow \text{ (amplifies cue reactivity)} \quad (15)$$

Net effect: χ drops rapidly. High stress compresses the adaptive basin.

7 Closed-Loop Control with Guardrails

In this section, $\chi(t)$ denotes the net adaptive susceptibility $\chi_{\text{net}}(t)$ unless otherwise specified.

7.1 Basic Control Law

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \dot{e}(t) \quad (16)$$

where $e(t) = \chi_{\text{target}} - \chi(t)$ and $\chi_{\text{target}} = 0.9$.

7.2 Guardrail Specifications

- **Deadband:** No intervention for $0.85 < \chi < 0.95$ (avoid overtreatment)
- **Rate limiting:** $|\dot{u}| < u_{\text{max}}$ (prevent oscillatory overshoot)
- **Anti-windup:** Clamp integrator when χ saturates at bounds
- **Multi-mode balancing:** Weight interventions across all χ_i

7.3 Decision Rules

$$\text{Action} = \begin{cases} \text{Intensify } \gamma\text{-restorer} & \text{if } \chi < 0.75 \text{ and } \dot{\chi} < 0 \\ \text{Intensify } \omega\text{-suppressor} & \text{if } \chi < 0.75 \text{ and } \dot{\chi} \geq 0 \\ \text{Maintain regimen} & \text{if } 0.75 \leq \chi \leq 1.0 \\ \text{Reduce damping} & \text{if } \chi > 1.2 \text{ (overdamped)} \\ \text{Emergency protocol} & \text{if } \chi < 0.5 \text{ or } \kappa > 1.0 \end{cases} \quad (17)$$

8 Medication-Assisted Treatment as χ -Modulation

Table 3: MAT mechanisms in SymC framework

Medication	Class	ω Effect	γ Effect	χ Result
Methadone	Full agonist	Steady floor	Indirect stabilization	Smooths variance
Buprenorphine	Partial agonist	Ceiling effect	Mild increase	Prevents crashes
Naltrexone	Antagonist	Blocks spikes	Forces recovery	Enables γ repair
Acamprosate	GABA/Glu mod.	Reduces drive	Increases damping	Dual stabilization
Disulfiram	Aversive	Adds negative ω	—	Behavioral deterrent

9 Clinical Trial Protocol

9.1 Phase 0: Feasibility

- $N = 30$ patients across addiction types
- Collect EEG, HRV, craving diaries for 4 weeks
- Extract χ_{reward} , χ_{control} , χ_{stress}
- Validate estimator stability (test-retest $r > 0.7$)

9.2 Phase 1: Stratification Validation

- $N = 100$ patients
- Classify into Types A–D based on χ signatures
- Test subtype prediction of treatment response
- Primary endpoint: Classification accuracy vs. clinical judgment

9.3 Phase 2: χ -Guided Therapy

- $N = 200$ patients, randomized
- Arm A: Standard treatment
- Arm B: χ -targeted treatment (type-matched interventions)
- Primary endpoint: 6-month relapse rate
- Secondary: Time to relapse, χ -stabilization time, quality of life

9.4 Phase 3: Closed-Loop Adaptive Control

- $N = 50$ patients with wearable χ -monitoring
- Real-time intervention adjustment based on $\chi(t)$
- Compare to fixed-protocol arm
- Primary endpoint: Sustained residence in adaptive window

10 Extended Falsification Matrix

#	Prediction	Test Method	Falsification Criterion
1	$\chi_{\text{reward}} - \chi_{\text{control}}$ precedes relapse	Longitudinal EEG monitoring	No temporal precedence
2	Recovery follows U-curve	Serial χ measurement	Monotonic or random trajectory
3	Withdrawal requires $\chi > 1.2$	χ during acute withdrawal	Anhedonia with $\chi < 1.0$
4	Type C shows context-dependent χ	Environment-switching paradigm	No context effect on χ
5	MAT reduces Q-factor	Spectral analysis pre/post MAT	No bandwidth change
6	High-Q in addicted vs. control	Cue-evoked EEG comparison	No Q-factor difference
7	χ -guided therapy outperforms	RCT comparison	No outcome difference
8	Curvature κ predicts relapse	κ vs. relapse timing	χ -value outperforms κ

11 Cross-Domain Comparison

Table 5: SymC boundary across domains

Domain	Underdamped ($\chi < 1$)	Overdamped ($\chi > 1$)	Key Failure
Addiction	Craving, compulsion	Withdrawal, anhedonia	Bandwidth collapse
Dementia	Early instability	Late rigidity	Stiffness cascade
Parkinson’s	Tremor	Rigidity, freezing	Controller failure
Chronic Pain	Central sensitization	Numbness	γ -collapse
Oncology	Relapse oscillation	Therapy rigidity	Adaptive control
Seismology	Fault instability	Locked fault	Impedance mismatch

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